留嘉城 Jiacheng Liu

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教育背景

上海交通大学 · 人工智能学院 2026.09 – 至今
计算机科学与技术 · 博士研究生；指导教师：张林峰教授。

山东大学 · 机电与信息工程学院 2022.09 – 2026.07
计算机科学与技术 · 本科。

研究方向

Coding Agent 任务构建、长程信用分配、Multi-turn agent 异步训练、On-Policy Distillation 与跨模型知识迁移；高效生成模型与交互式游戏世界模型。

实习经历

无尽前沿（Endless Frontier） 2026.06 – 至今
LLM 基模组 · 后训练实习生 · 上海

腾讯混元 · 多模态模型部 2025.07 – 2026.06
世界模型基模组 · 青云计划研究实习生 · 上海

大语言模型研究经历

1. Coding Agent：任务生成与分布式验证

构建从仓库与 PR/Issue 采集、候选筛选、版本定位到容器化任务生成与验证的数据流水线，产出包含题面、修复前代码、参考补丁及测试的可执行任务。基于 Temporal 编排多 Worker 并行执行，支持全局资源配额、失败重试和已提交结果复用；通过隔离测试与修复前后对照，校验任务可解性和判分有效性。

2. 长程 Agent：三角色异步训练与信用分配

面向 27B→9B Agent Coding 任务，实现 Rollout、Teacher、Actor 三角色轮级异步流水线。通过教师前缀缓存、轨迹内上下文合并和完整轨迹提前反向衔接评分与训练，在有效 Token 窗口内累积梯度、统一更新并发布权重。冻结教师骨干，以学生完整轨迹的终局成败离线训练前缀价值头，评分时共享教师前向；根据相邻 Token 前缀的价值差构造非负权重，对 Actor 的 OPD 损失进行加权。

3. 能力迁移：跨 Tokenizer 蒸馏与多教师融合

在共享字节空间中对齐师生预测，将教师概率映射为学生词表上的监督目标；针对教师事件跨越多个学生 Token 的情况，聚合合法补全 Token 的概率，监督后续生成。在 Kimi 1T→Qwen 35B、Qwen 397B→30B 的跨 Tokenizer MoE 蒸馏中验证；其中 Kimi 配置在物理推理基准 PHYRD-40 上的平均分较最强对比基线提升 5.1 分。同时通过 MOPD 融合数学、代码与指令遵循能力，并参与工业代码及多模态任务的 27B 级模型合版。

以上代表工作包括 ESCD（ICLR 在投）、TRAIL、PHYRD（ICLR 在投）与 BigBang（技术报告）。

多模态研究经历

1. 高效生成模型：特征预测与自适应计算

围绕扩散模型的重复计算，研究基于历史特征演化的预测式缓存，将直接复用拓展为特征预测，并探索预测校准、阶段感知预测器、自适应计算与频域缓存，覆盖图像、视频、编辑及语音生成。代表工作包括 TaylorSeer（ICCV 2025）、SpeCa（ACM MM 2025）、LESA（CVPR 2026）与 FreqCa（NeurIPS 2026）。

2. 交互式游戏世界模型：Hunyuan-GameCraft-2

在腾讯混元世界模型基模组参与 Hunyuan-GameCraft-2 研发，面向自然语言指令与玩家动作驱动的游戏画面生成。基于 Unity 渲染游戏场景视频，为世界模型构建训练数据，并开展 3D 相关基线实验；参与交互视频清洗与标注、交互记忆教师模型 SFT、Self-Forcing / DMD 少步蒸馏、强化学习优化与评测，关注动作响应与长时序生成一致性。相关成果包括 Hunyuan-GameCraft-2（技术报告、CVPR 2026 Workshop）。

论文与技术报告

粗体作者为本人；* 表示同等贡献，† 表示通讯作者。

[ICLR 2027 Under Review] Beyond Token Alignment: Event Completion for Cross-Tokenizer On-Policy Distillation. **Jiacheng Liu***, Song, J.*, Zhang, Q., Chen, S., Linfeng Zhang†. [论文] [代码] [报告] [SciDeriv] [PHYRD-40]
[ICLR 2027 Under Review] RolloutRemover: Scribble-Guided Object Removal via Latent Rollout Modeling. Wang, Z., Li, Y., **Jiacheng Liu**, Ma, Y., Huang, W., Liu, Q., Zhang, J.†.

[ICLR 2027 Under Review] TempoKV: Predictive KV Cache Quantization For Autoregressive Video Generation. Hu, J., Mo, W., Cai, P., **Jiacheng Liu**, Ren, J., Zhang, X., Tu, X., Linfeng Zhang†.
[ICLR 2027 Under Review] Search Low, Refine the Winner: Fast Inference through Noise-Trajectory Selection. Qin, H., Yan, Z., **Jiacheng Liu**, Peng, H., Liu, J., Tu, X., Ren, J., Zhang, X., Wang, Y., Linfeng Zhang†.
[ICLR 2027 Under Review] PHYRD: A Living Benchmark for Reproducing and Extending Theoretical Physics Research. Zhang, Q., Chen, G., Zhang, W., Wang, J., Song, J., **Jiacheng Liu**, Chen, S., Linfeng Zhang†.
[Neurocomputing Under Review] PCMamba: Parallel Convolution-Mamba Network for Medical Image Segmentation. **Jiacheng Liu**, Dong, L., Chen, B., Shi, J., Lu, W., Zhang, S., Yang, F.†, Li, D., & Zhang, L. [论文] [代码]
[TMLR 2026] A Survey of Linear Attention: Algorithm, Theory, Application, and Infrastructure. Zhang, Y., Sun, W., Sun, S., Ding, W., Xie, R., Wang, H., Chen, J., **Jiacheng Liu**, Wang, S., Zhang, Y., Huang, Y., Wang, J., Zhao, T., Han, W., Chen, Y., Zhang, K., Li, S., Xie, R., Wang, D., Chen, J., Zhang, L., Xu, C., & Wang, Y.†. [论文] [资源]
[NeurIPS 2026] Focused Forcing: Content-Aware Per-Frame KV Selection for Efficient Autoregressive Video Diffusion. Cai, P., Zhang, E., **Jiacheng Liu**, Lin, H., Zhang, R., Mo, W., Ma, Y., Zheng, S., Huang, J., Liu, D., & Zhang, L.†. [论文]
[NeurIPS 2026] Dynamic Video Generation: Shaping Video Generation Across Time and Space. Zheng, S., Huang, J., **Jiacheng Liu**, Chen, G., He, L., Lin, Y., Cai, P., & Zhang, L.†. [论文] [代码]
[NeurIPS 2026] FreqCa: Accelerating Image Generation and Editing via Frequency-Aware Caching. **Jiacheng Liu**, Cai, P., Zhou, Q., Lin, Y., Kong, D., Huang, B., Pan, Y., Xu, H., Zou, C., Tang, J., Zheng, S., & Zhang, L.†. [论文] [代码]
[NeurIPS 2026] Z-Cache: Accelerating Diffusion Transformers via Self-Reflection. Cheng, Z.*, Wang, Z.*, **Jiacheng Liu**, Tu, X., Zhou, C., Chen, J., Ma, Y., Ma, Z., Ren, J., Zou, L., & Zhang, L.†.
[ECCV 2026] LinCa: Accelerating Diffusion Models via Learnable Decomposed Feature Caching. Liu, J.*, Qin, H.*, Tu, X., **Jiacheng Liu**, Hu, J., Yan, Z., Xie, Y., Shen, K., Ren, J., Lin, Y., Zhang, X., & Zhang, L.†. [论文] [代码] [权重]
[ECCV 2026] AViTS: Adaptive Spatiotemporal Token Selection for Efficient Dynamic-Resolution Generation. Qin, H.*, Yan, Z.*, Zheng, S.*, Tu, X., **Jiacheng Liu**, Lin, Y., Zou, C., Liu, J., Cai, P., Zhang, X., Ren, J., & Zhang, L.†. [论文]
[ECCV 2026] Accelerating Diffusion Transformers with Gaussian Process Rectified Feature Cache. Shen, Z.*, Huang, R.*, Zou, C., Zheng, S., **Jiacheng Liu**, Cai, P., Shi, Z., Du, Y., Feng, L., Tu, X., Ren, J., Zhang, X., & Zhang, L.†. [论文]
[CVPR 2026 Workshop · Technical Report] Hunyuan-GameCraft-2: Instruction-following Interactive Game World Model. Junshu Tang*, **Jiacheng Liu***, Jiaqi Li*, Longhuang Wu, Haoyu Yang, Penghao Zhao, Siruis Gong, Xiang Yuan, Shuai Shao, Linfeng Zhang, Qinglin Lu†. [论文] [项目]
[Technical Report 2026] BigBang: Pursuing Open-Ended Intelligence through Self-Evolving Synthesis of Verifiable Frontier Tasks. The BigBang Team. [论文] [项目] [权重]
[进行中] TRAIL: Critic-Guided On-Policy Distillation for Multi-Turn Agents. **Jiacheng Liu***, Jingwei Song*, Siheng Chen, Linfeng Zhang†.
[BSPC 2025] FQCDM: Feature Quantization-Based Cardiac Image Diffusion Synthesis Model. Shi, J., Chen, B., **Jiacheng Liu**, Lu, W., Zhang, S., Ma, W., Yang, F.†, & Li, D. [论文]
[CVPR 2026] LESA: Learnable Stage-Aware Predictors for Diffusion Model Acceleration. Peiliang Cai*, **Jiacheng Liu***, Haowen Xu, Xinyu Wang, Chang Zou, Linfeng Zhang†. [论文] [代码]
[CVPR 2026] From Sketch to Fresco: Efficient Diffusion Transformer with Progressive Resolution. Zheng, S., Chen, G., He, L., **Jiacheng Liu**, Lin, Y., Zou, C., & Zhang, L.†. [论文]
[CVPR 2026] Beyond Fixed Formulas: Data-Driven Linear Predictor for Efficient Diffusion Models. Shen, Z.*, Huang, R.*, **Jiacheng Liu**, Zou, C., Cai, P., Zheng, S., Shi, Z., Feng, L., & Zhang, L.†. [论文] [代码]
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[ICLR 2026 Oral] Let Features Decide Their Own Solvers: Hybrid Feature Caching for Diffusion Transformers. Zheng, S., Chen, G., Zhou, Q., Lin, Y., He, L., Zou, C., Cai, P., **Jiacheng Liu**, & Zhang, L.†. [论文]
[ICLR 2026] HiCache: Training-free Acceleration of Diffusion Models via Hermite Polynomial-based Feature Caching. Feng, L.*, Zheng, S.*, **Jiacheng Liu**, Wang, X., Zou, C., Ma, Y., Chen, J., Cai, P., & Zhang, L.†. [论文] [代码]
[AAAI 2026] WaveEx: Accelerating Flow Matching-based Speech Generation via Wavelet-guided Extrapolation. Liu, X.*, Gui, X.*, Ge, Z., Ge, Y., Zou, C., **Jiacheng Liu**, Niu, Z., Zheng, Q., Xu, C., Chen, X., Xiao, T., Zhu, J., & Zhang, L.†. [论文]
[AAAI 2026] Forecast then Calibrate: Feature Caching as ODE for Efficient Diffusion Transformers. Zheng, S., Feng, L., Wang, X., **Jiacheng Liu**, Zou, C., Ma, Y., Chen, J., Cai, P., & Zhang, L.†. [论文] [代码]
[ACM MM 2025] SpeCa: Accelerating Diffusion Transformers with Speculative Feature Caching. **Jiacheng Liu***, Chang Zou*, Yuanhuiyi Lyu, Fei Ren, Shaobo Wang, Kaixin Li, Linfeng Zhang†. [论文] [代码] [项目]
[BSPC 2025] PSVT: Pyramid Shifted Window based Vision Transformer for Cardiac Image Segmentation. Zhang, X., **Jiacheng Liu**, Xian, X., Chen, B., Li, D., Yang, F.†, & Zhang, L. [论文] [代码]
[ICCV 2025] From Reusing to Forecasting: Accelerating Diffusion Models with TaylorSeers. **Jiacheng Liu***, Chang Zou*, Yuanhuiyi Lyu, Junjie Chen, Linfeng Zhang†. [论文] [代码] [项目]
[Preprint 2025] Accelerate Diffusion Transformers with Feature Momentum. Fang, J., Zou, C., **Jiacheng Liu**, Lyu, Y., Hu, X., & Zhang, L.†. [论文]

奖项与荣誉

创业实践类与研究创新类特长奖一等奖、优秀学生一等奖、校三好学生。